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White-Box Adversarial Exploitation of NIDS: Insights from FGSM, PGD, and C&W

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***Abstract*— Network Intrusion Detection Systems (NIDS) help in detecting harmful activities and malicious activities by keeping an eye on network traffic or system for malignant activity. However, studies have found that Network Intrusion Detection Systems (NIDS) can be tricked by making tiny often invisible changes to the input data known as adversarial attacks to confuse machine learning models and cause them to make wrong decisions. This paper explores how different types of adversarial attacks affect NIDS models, focusing on techniques like Fast Gradient sign Method (FGSM), Projected Gradient Descent (PGD), and the Carlini and Wagner (C&W) attack. This work contributes by assessing how these adversarial attacks affect the NIDS model by analyzing three key metrics: precision, recall, and F1 score. These metrics were analyzed to evaluate how successful the adversarial attacks were and how they affected the performance of the NIDS. Experimental results demonstrate that these adversarial attacks significantly degrade the performance of NIDS, particularly in terms of precision and overall detection capability. FGSM was able to reduce the accuracy of NIDS from 73.01% to 53.92% as epsilon (ϵ) value increased from 0.1 to 0.5, while PGD was able to reduce it to 19.63%. The C&W attack, with the confidence parameter set to 1, resulted in 55.20% accuracy, highlighting the model's vulnerability to adversarial attacks. This research highlights the vulnerabilities of NIDS to adversarial attacks and provides a detailed analysis of the challenges posed by such threats in cybersecurity, emphasizing the need for advanced mitigation techniques to address them.**

Keywords*— Network Intrusion Detection Systems (NIDS), adversarial attacks, CNN-LSTM, FGSM, PGD, Carlini & Wagner attack, robustness, network security.

I. INTRODUCTION

In the scope of cybersecurity, Network Intrusion Detection Systems (NIDS) are essential for protecting networks from harmful activities. These systems analyse network traffic in order to detect and appropriately respond to any potential threats. As cyber threats become more complex, these systems are increasingly under threat. One big challenge is that they can be tricked by adversarial attacks. These happen when small changes are made to the data, which causes the system's machine learning models to make mistakes, and it then makes incorrect classifications.

Adversarial attacks pose a major challenge to NIDS, as they

can trick systems into misclassifying malicious activities. Attacks such as Project Gradient Descent (PGD), Fast Gradient Sign Method (FGSM), and the Carlini & Wagner (C&W) have been shown to exploit weaknesses which involves altering network traffic in a way that is not easily detectable but still capable of evading traditional detection mechanisms. Such exploits highlights the need for robust defensive strategies.

This study focuses on understanding the influence of these adversarial attacks on a CNN-LSTM-based NIDS model, using the CICIDS 2017 dataset. The CNN-LSTM architecture was selected over other architectures like GRU or purely CNN based models because of its proven ability to capture both spatial patterns (via convolutional layers) and temporal dependencies (via LSTM layers), making it well-suited for analyzing network traffic data. White-Box adversarial attacks such as FGSM, PGD and C&W were selected as they represent the worst-case scenario where the attacker has complete knowledge about the model's architecture, parameters and gradients that allowed for a thorough analysis of the model's robustness. In this study, we examined the impact of three major white-box adversarial attack techniques—FGSM, PGD, and C&W on deep learning-based Network Intrusion Detection Systems (NIDS).

The overall research is organized as follows: Section II discusses the background study, and Section III explains the methodology adopted. Section IV discusses the experimental results. Section V discusses the challenges faced and Section VI concludes the overall research work.

II. BACKGROUND STUDY

A. Growth of Network Intrusion Detection systems

Network Intrusion Detection Systems (NIDS) have grown significantly over time. Early systems were dependent upon signature-based methods that compared network traffic to predefined attack patterns. While effective for detecting known threats, these approaches were limited in their ability to identify new or unknown attacks. The growth of machine learning has transformed NIDS by helping them to detect complex and emerging attack behaviours by learning patterns from historical data. This advancement has made NIDS more adaptable and dynamic in addressing the evolving landscape of cyber threats.

B. Machine learning based NIDS vulnerabilities

The integration of machine learning into NIDS has many benefits but it has also introduced a critical vulnerability:

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susceptibility to adversarial attacks. Adversarial attacks makes unnoticeable changes to the input data thus, taking advantage of weaknesses in machine learning based NIDS models. These subtle modifications can lead the system to incorrectly label malicious traffic as harmless also known as benign. This is especially troubling because such attacks overcome the usual strengths of machine learning, creating a serious challenge for the effectiveness of NIDS.

C. State of Research on Adversarial NIDS

As NIDS models evolve, they also become vulnerable to adversarial attacks, which exploit weaknesses in the model to bypass detection and cause misclassification of harmful traffic. The adversarial machine learning is a critical area of research as these attacks can lead to serious security risks for organizations that depend on intrusion detection systems for protection.

1) Overview of Adversarial Attacks on NIDS

There are various sorts of adversarial attacks on NIDS, but the most prominent ones are the Projected Gradient Descent (PGD), Fast Gradient Sign Method (FGSM), and Carlini & Wagner (C&W) attack. These techniques are intended to alter input data in ways that are frequently undetectable and unnoticeable to humans but result in incorrect input classification by the machine learning model. Each of these attacks has been studied in depth for their ability to bypass modern NIDS.

- Fast Gradient Sign Method (FGSM):

FGSM is a simple yet powerful technique put forwarded by Goodfellow *et al.* [1] which makes use of the gradients taken from the model's loss function to create adversarial samples in a single step. It is computationally efficient and effective for quick evaluations of model robustness [3]. FGSM uses loss function's gradient with respect to the input in order to create adversarial samples. Let p be the input data, q the true label, and $J(\omega, p, q)$ the loss function of the model with parameters ω. The gradient represented as ∇_p is computed by the formula given in (1)

$$\nabla_p J(\omega, p, q) = \partial p / \partial J \quad (1)$$

The perturbation δ is computed and shown in (2)

$$\delta = \epsilon \cdot sign(\nabla_p J(\theta, p, q)) \quad (2)$$

where ϵ is the constant controlling the magnitude of the perturbation.

The adversarial example x_{ae} is created by adding the perturbation to the original given in (3)

$$x_a e = x + \sigma \quad (3)$$

2) Projected Gradient Descent (PGD):

The PGD attack was proposed by Madry *et al.* [2] and is widely used for evaluating adversarial robustness. PGD is an iterative version of FGSM where a number of small perturbations are applied to input data over several steps. This iterative process enhances the attack's success rate. In the context of NIDS, PGD has proven to be effective at bypassing defences, including those based on adversarial training, making it a significant threat to the robustness of detection systems. This process represented by (4).

$$\theta_{t+1} = \theta_t - \alpha \cdot AJ(\theta_t) \quad (4)$$

Where θ_t denotes the parameters at iteration t, α is the learning rate, and $AJ(\theta_t)$ represents the gradient of the loss function.

Projected Gradient Descent (PGD) extends this concept by introducing a perturbation budget (ϵ) and a step size (α), which govern the magnitude and direction of the perturbation applied to the input. The iterative update rule for PGD is given in (5)

$$pt+1' = \Pi(pt+\alpha \cdot sign(ApJ(\theta, pt, q))) \quad (5)$$

where pt is the input at iteration t, α denotes the step size, $\nabla_p J(\theta, pt, q)$ is the gradient of the loss function with respect to the input, and Π represents the projection operator and it ensures that the perturbed input remains within the predefined bounds.

- Carlini & Wagner Attack(C&W):

The C&W attack minimizes the perturbation while maintaining the adversarial sample's effectiveness. This method is particularly challenging to detect as it introduces minimal changes to the original input data. The C&W attack is particularly effective against NIDS because it can bypass traditional defense mechanisms like adversarial training, making it a powerful tool in adversarial machine learning research [4].

The C&W attack minimizes the distance between the original input x and the adversarial example x_{adv} while ensuring the adversarial example tricks the NIDS model. The optimization problem is formulated as given in (6)

$$\min \|\delta\|_p + c \cdot f(x + \delta) \quad (6)$$

The optimization problem is then subjected to the problem given in (7)

$$x_{adv} = x + \delta \text{ and } x_{adv} \in [0,1]^n \quad (7)$$

Here:

a) δ: The perturbation added to the input.

b) $\| \delta \|_p$: The p -norm of the perturbation, typically L_2.

c) f(x+δ): An objective function designed to ensure misclassification.

d) c: A constant balancing the trade-off between minimizing ‖δ‖ and maximizing the attack's effectiveness.

D. Challenges and Solutions in Defending Against Adversarial Attacks on NIDS

NIDS are vulnerable to adversarial attacks which pose a significant challenge as machine learning models become more widely used for intrusion detection. Several defence strategies have been proposed but each of these strategies have their own limitations:

- Adversarial Training: This involves training the model with both normal and adversarial examples to improve its robustness. This strategy is effective but it is computationally expensive and it may decrease performance on non-adversarial inputs [1].
- Feature Manipulation: This method alters input features to reduce the impact of adversarial

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perturbations. This strategy is promising but it is still being researched, especially for handling real-time network traffic [4].

- Ensemble Methods: By combining the predictions of multiple models, ensemble methods can make it harder for attacks to bypass detection. However, they add complexity and increase computational demands [3].

III. METHODOLOGY

The NIDS model used is a hybrid deep learning model designed for analyzing sequential data. The model is explained by its schematic diagram as shown in Fig. 1. It begins with an input layer, where raw data is passed through convolutional filters (Conv1D) to identify key features and then it is passed through pooling layers to reduce complexity and focus on essential patterns. Batch normalization ensures stable learning throughout the process. LSTM layers then capture time-based relationships, allowing the model to understand sequences effectively. Finally, the data is flattened and passed through dense and dropout layers to extract meaningful information, prevent overfitting and make accurate predictions through the output layer. The chosen hyperparameters were optimized based on empirical performance. A dropout rate of 0.5 was applied to the fully connected layers to prevent overfitting. The learning rate was kept at Adam optimizer setting to ensure stable and efficient convergence. A batch size of 64 epochs was selected to balance between training efficiency and model performance.

A. Dataset Description

The CICIDS 2017 [5] dataset was used in this study to develop and evaluate a Network Intrusion Detection System (NIDS). The dataset contains traffic traces from normal and attack scenarios, including distributed denial-of-service (DDoS), brute force, botnet, and port scanning attacks. After preprocessing, the dataset consisted of approximately 223,000 samples, with 78 features representing network traffic characteristics.

B. Data Preprocessing

The dataset underwent the following preprocessing steps:

1) Loading the Data: The preprocessing of the data was done to remove missing and infinite values. Missing numeric values were replaced with column means, and infinite values were replaced with medians.

2) Handling Class Imbalance: To address the issue of class imbalance in the dataset which could lead to the model performing poorly on the minority classes, Synthetic Minority Over-sampling Technique (SMOTE) was applied to the training data. SMOTE generates synthetic samples for the minority classes which ensures that the model receives a more balanced representation of all classes during training.

3) Feature Engineering: Label encoding was applied to convert categorical labels into numerical form. The target column (Label) was encoded using a LabelEncoder, transforming class names into integer labels.

4) Feature Scaling: StandardScalar was used to normalize the feature values and to standardize data for model training.

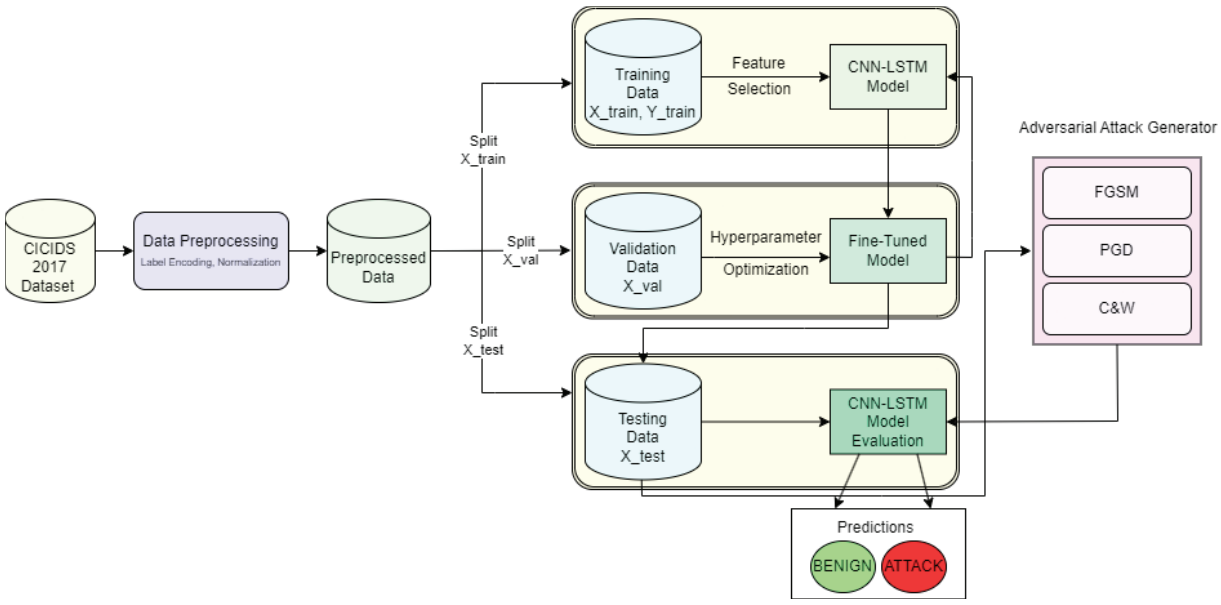


Fig. 1. Schematic Diagram for Network Intrusion Detection System Development

- Reshaping for Temporal Dependencies: Input features were reshaped into sequences with dimensions (samples, time_steps, features).

The dataset was segmented such that the number of features (78) was divided into **13-time steps**, resulting in 6 features per time step.

C. Model Architecture

A CNN-LSTM hybrid architecture was chosen whose structure is defined in Fig. 2. It was chosen to take advantage of the following properties :

1. CNN for feature extraction.
2. LSTM for capturing temporal dependencies.

Model Layers:

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1) Convolutional Layers:

- First Conv1D: 64 filters, kernel size 3, activation function ReLU.
- Second Conv1D: 128 filters, kernel size 3, activation function ReLU.
- Each convolutional block was followed by:
- Batch Normalization to stabilize training.
- MaxPooling1D(pool_size=2) to reduce dimensionality.

2) LSTM Layers:

- First LSTM: 100 units, ret_seq=True, recurrent_dropout=0.2
- Second LSTM: 100 units, ret_seq=False, recurrent_dropout=0.2
- Output Layer: Number of nodes equal to the number of classes, softmax activation function.

3) Fully Connected Layers:

- Flattened the LSTM output.
- Dense Layer 1: 128 nodes, ReLU activation, with a Dropout (0.5) layer.
- Dense Layer 2: 64 nodes, ReLU activation, with a Dropout (0.5) layer.

1. Compilation:

- Input Shape: (13, 6) for each time-series instance.
- Optimizer: Adam
- Loss Function: Sparse Categorical Crossentropy

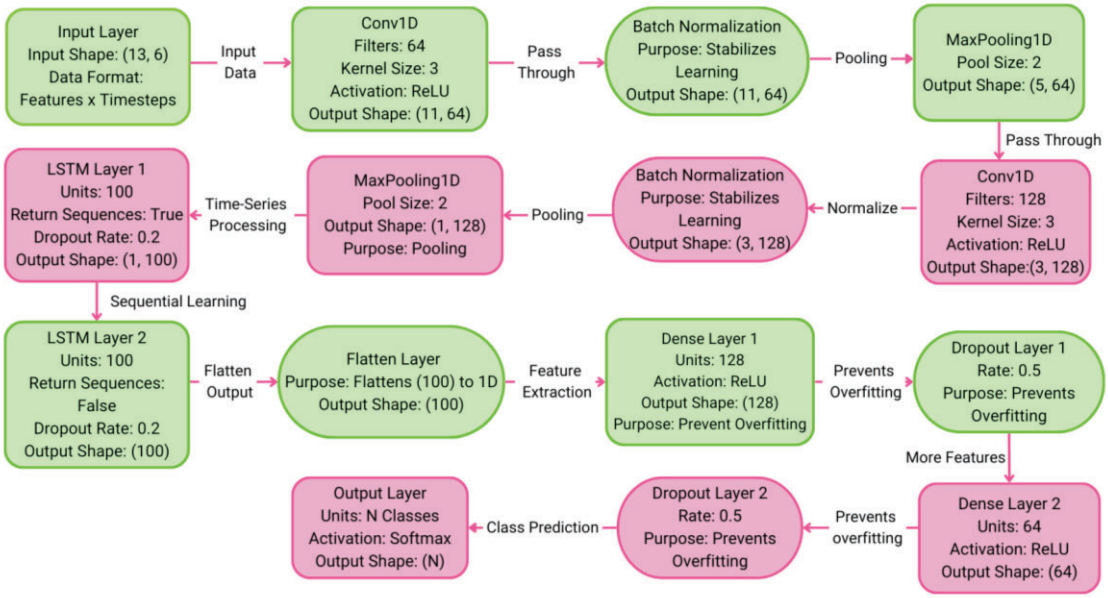


Fig. 2. Model Structure of the CNN-LSTM based NIDS

D. Training

- The dataset was partitioned into three subsets: 60% for training, 20% for validation, and 20% for testing. Sampling techniques were employed to ensure balanced class representation across all subsets which preserved the distribution of classes for reliable model evaluation.
- Early Stopping: To prevent overfitting, training was monitored with a patience of 5 epochs using validation loss.
- Hyperparameters: Epochs: 30, Batch Size: 64, Optimizer learning rate: Adam settings.

1) Evaluation Metrics

The performance and efficiency of the CNN-LSTM model is assessed using several evaluation metrics, with a particular focus on the model's capability to accurately classify network traffic:

- Accuracy: Accuracy [6] was used as a primary metric, indicating the ratio of correctly classified instances (including both normal and attack traffic) to the total number of instances.
- Precision: The percentage of true positives among all predicted positives, indicating how accurate the model is in identifying network attacks [6].

3. Recall: The percentage of true positives among all actual positives, measuring the model's ability to correctly identify attacks.
4. F1-Score: The harmonic mean of precision and recall which balances both false positives and false negatives.
5. Support: The number of actual occurrences of each class in the test dataset which helps to understand how well the model handles class imbalances.

E. Adversarial Attacks

To evaluate the resilience and robustness of the CNN-LSTM model, three adversarial attacks were applied: FGSM, PGD, and C&W. The results after the application of these attacks on the NIDS model which includes the parameters used and the model's performance, are summarized in Table I.

6. FGSM (Fast Gradient Sign Method):

- Parameters: epsilon = 0.1,0.2,0.3,0.4,0.5

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- Adversarial examples for FGSM were generated by adding perturbations proportional to the gradient of the loss function.

7. PGD (PROJECTED GRADIENT DESCENT):

- Parameters: epsilon = 0.1,0.2,0.3,0.4,0.5
- step size = 0.01, iterations = 40

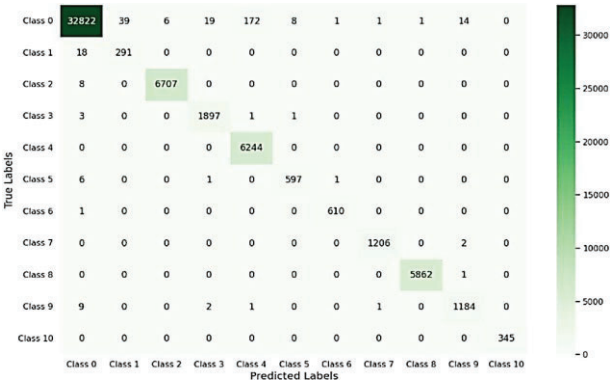


Fig. 3. Confusion Matrix Heatmap for Original Input Data

PGD is an iterative method that perturbs inputs to maximize classification error.

8. C&W (Carlini & Wagner):

- Parameters: Confidence = 1, Learning Rate = 0.01, Iterations = 100
- It is a targeted attack that is optimized to minimize the distortion while fooling the classifier.

The CNN-LSTM model performed well on the CICIDS 2017 dataset with high accuracy. Adversarial attacks exposed vulnerabilities, highlighting the need for adversarial training as future work.

IV. RESULT AND ANALYSIS

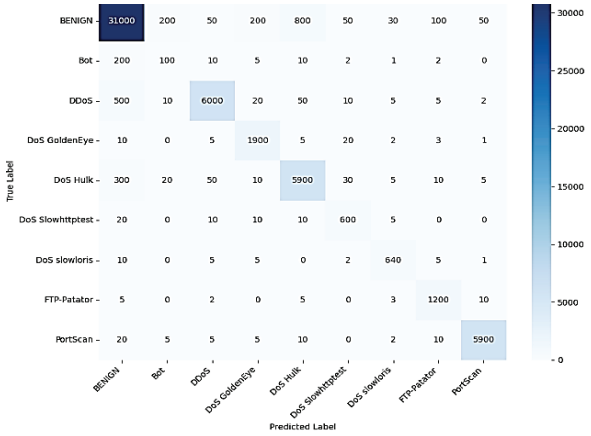


Fig. 4. Confusion Matrix Heatmap after FGSM attack

This section presents and analyses the functionality and effectiveness of the CNN-LSTM network intrusion detection system (NIDS) model. The Model performs well with high test accuracy on original input data without any perturbations. When perturbations were introduced in the form of FGSM, PGD and C&W attack the model misclassified harmful traffic as benign which can be observed by a significant drop in the model's accuracy.

A. Model Performance on Clean Data

The CNN-LSTM model demonstrated exceptional performance on the clean CICIDS 2017 dataset. The model achieved a high-test accuracy of 99.45%, indicating its ability to detect network intrusions reliably.

B. Confusion Matrix Analysis

The confusion matrix for original input data (Fig. 3) illustrates the model's excellent classification capability as it identified most samples correctly. However, the confusion matrices for adversarial data (Fig. 4) reveal an increase in misclassifications after the introduction of FGSM attack.

TABLE I. ADVERSARAL ATTACK SUMMARY

Attack Type	Parameter	Accuracy (%)
FGSM	Epsilon (ϵ)	73.01, 64.24, 59.10, 56.03, 53.92
PGD	Epsilon (ϵ), Iterations (n)	36.60, 23.68, 20.08, 19.65, 19.63
C&W	Confidence (c)	55.20

C. Impact of Adversarial Attacks

To evaluate robustness, the model was subjected to three adversarial attacks: FGSM, PGD, and C&W. Table II summarizes the results which showcased the model's performance degradation under each attack for epsilon(∈) = 0.1. The CNN-LSTM model showcased excellent performance on clean data thus, validating the effectiveness of its architecture for detecting network intrusion. The results highlights and stresses on the importance of incorporating adversarial training or other defense mechanisms to improve resilience. To provide a clearer understanding of the impact of adversarial attacks, Fig. 5 displays a bar chart comparing the model's accuracy on clean data versus adversarial samples and Fig. 6 displays how different classes were classified by the model after being subjecting to different attacks.

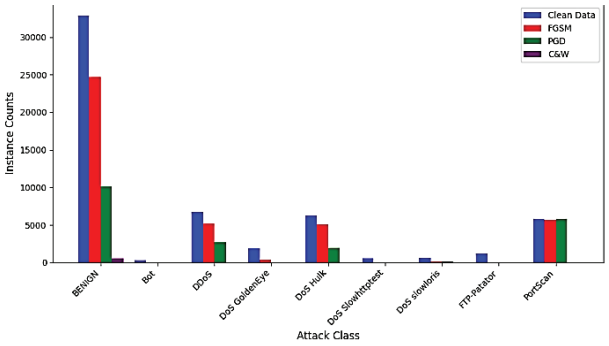


Fig. 5. Comparison of Attack Types

V. CHALLENGES FACED

The CNN-LSTM architecture demonstrated its effectiveness in detecting network intrusions. The convolutional layers efficiently captured patterns in the data, while the LSTM layers processed sequential dependencies, enabling the model to excel in identifying attack behaviours resulting to high accuracy on clean data, validating this choice of architecture for network intrusion detection systems (NIDS). Despite the promising results,

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the model's robustness and resilience to adversarial attacks revealed a critical vulnerability. The FGSM, PGD, and C&W attacks significantly impacted the classification accuracy, with the FGSM attack proving the most effective at misleading the model.

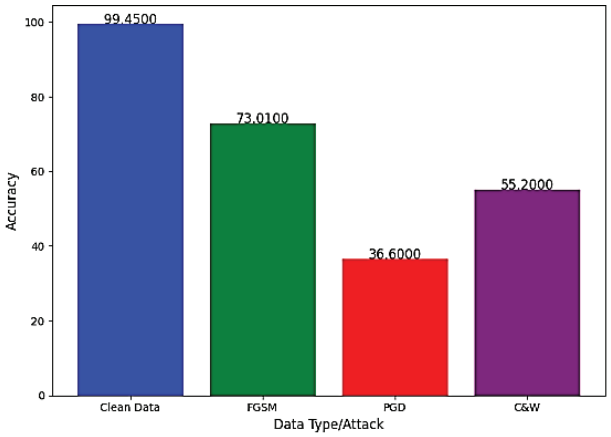


Fig. 6. Model Accuracy on Clean and Adversarial Data

VI. CONCLUSION

The primary objective of this research was to develop and evaluate a CNN-LSTM-based model for network intrusion detection (NIDS) using the CICIDS 2017 dataset. The model aimed to classify network traffic data into multiple categories, enabling the identification of various intrusions. This study highlights critical vulnerabilities in NIDS models that have high performance when exposed to clean data but when subjected to adversarial attacks have notable decline in F1 score, precision, and recall under such conditions. The experiments demonstrated that the NIDS model achieved an accuracy of 99.45% on clean data. However, adversarial attacks such as PGD, C&W, and FGSM posed significant challenges, reducing accuracy to 36.60%, 55.20% and 73.01% respectively. The findings align with growing concerns about adversarial robustness in machine learning models, indicating that advanced defenses or training techniques could enhance performance and that there is a need for industry practitioners to evaluate and harden their systems against such threats to maintain robust cybersecurity.

TABLE II. PERFORMANCE OF NIDS MODEL AGAINST ADVERSARIAL ATTACKS

Metric	Clean Data (%)	FGSM (%)	PGD (%)	C&W (%)
Accuracy	99.45	73.01	36.60	55.20
Precision	98.90	41.90	25.55	42.49
Recall	99.20	42.72	22.51	55.20
F1-Score	99.05	41.89	23.27	48.02

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